

Chaitanya Amballa

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PhD Student, ECE · University of Illinois Urbana–Champaign

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Scholar: [Google Scholar](#)

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Website: achaitanya.web.illinois.edu

Education

University of Illinois Urbana–Champaign

Urbana, IL

Ph.D. in Electrical and Computer Engineering

2022 – present

Advisor: Prof. Romit Roy Choudhury ([SiNRG Lab](#))

Research: generative modeling, spatial inference, diffusion models for inverse problems

University of Illinois Urbana–Champaign

Urbana, IL

M.S. in Electrical and Computer Engineering

2022 – 2025

Advisor: Prof. Romit Roy Choudhury

Indian Institute of Technology Tirupati

Tirupati, India

B.Tech. in Electrical Engineering

2015 – 2019

Advisor: Prof. K. P. Naveen

Honors & Awards

Qualcomm Graduate Award

2025

INSPIRE Scholarship

2015

Dept. of Science & Technology, India (top 0.1% in 12th standard exams)

2nd Prize, Intra IIT Table Tennis and Cricket

2018

IMMO Silver Medal

2013

NSO & NCO Gold Medals

2007, 2008

Sai Pratibha Puraskar

2013

Preprints

[1] **Contrastive Diffusion Guidance for Spatial Inverse Problems** [\[project page\]](#)



Sattwik Basu*, [Chaitanya Amballa](#)*, Zhongweiyang Xu, Jorge Vančo Sampedro, Srihari Nelakuditi, Romit Roy Choudhury

(* equal contribution)

[2] **Learning Energy-based Variational Latent Prior for VAEs**



Debottam Dutta, [Chaitanya Amballa](#), Zhongweiyang Xu, Yu-Lin Wei, Romit Roy Choudhury

[3] **Explicit Context-Driven Neural Acoustic Modeling for High-Fidelity RIR Generation**



Chen Si, Qianyi Wu, [Chaitanya Amballa](#), Romit Roy Choudhury

Publications

[1] **Inferring Indoor Layouts using Audio**



Zhijian Yang, Sattwik Basu, [Chaitanya Amballa](#), Debottam Dutta, Srihari Nelakuditi, Romit Roy Choudhury

HotMobile, 2026

[2] **Can NeRFs See without Cameras?** [\[project page\]](#)



[Chaitanya Amballa](#), Sattwik Basu, Yu-Lin Wei, Zhijian Yang, Mehmet Ergezer, Romit Roy Choudhury
NeurIPS, 2025

[3] **Identifying Near-Optimal Decisions in Linear-in-Parameter Bandit Models with Continuous Decision Sets**

Sanjay P. Bhat, [Chaitanya Amballa](#)

UAI, 2022



[4] **Computing an Efficient Exploration Basis for Learning with Univariate Polynomial Features**

[Chaitanya Amballa](#), Manu K. Gupta, Sanjay P. Bhat

AAAI, 2021



[5] **Coexistence of LTE-Unlicensed and WiFi: A Reinforcement Learning Framework**

K. P. Naveen, [Chaitanya Amballa](#)

COMSNETS, 2021



[6] **Learning Algorithms for Dynamic Pricing: A Comparative Study**

[Chaitanya Amballa](#), Narendhar Gugulothu, Manu K. Gupta, Sanjay P. Bhat

ICML Workshop on Real World Experiment Design and Active Learning, 2020



[7] **Coexistence of LTE-Unlicensed and WiFi: Optimization and Game-Theoretic Frameworks**

[Chaitanya Amballa](#), K. P. Naveen

SPCOM, 2020



Research
Projects

Human Body Pose Estimation with No Fine-Tuning [\[project page\]](#)

Oct 2025 – Dec 2025

Advisor: Dr. Zhizhen Zhao

Diffusion and flow matching for HMD-based pose estimation; posterior sampling for generalization across body shapes

RIR Generation and Geometry Estimation

Feb 2025 – May 2025

Advisor: Dr. Minje Kim

RIR generation using diffusion models; optimizing for unknown source position using Alexa measurements

Click2Hear: Spatial Audio Separation using Visual Cues [\[project page\]](#)

Aug 2022 – Dec 2022

Advisor: Dr. Saurabh Gupta

Unsupervised source separation; binaural audio from video with mono audio

Co-existence of LTE-Unlicensed and WiFi

Aug 2018 – May 2019

Advisor: Dr. K. P. Naveen (B.Tech Thesis)

Optimization, game-theoretic, and Q-learning frameworks for spectrum coexistence

Speech Enhancement

Aug 2018 – Dec 2018

Advisor: Dr. Rama Krishna Sai Gorthi

Convolutional autoencoder networks for real-time speech denoising

Blood Cell Classification and Disease Diagnosis

Sep 2017 – Jun 2018

Advisors: Dr. Rama Krishna Sai Gorthi & Dr. Subrahmanyam Gorthi

CNN-based classification of blood cells and disease detection from smear images

Industry Experience	Applied Scientist Intern	May 2025 – Aug 2025
	Amazon	Cambridge, MA
	Researcher	Jul 2019 – Jul 2022
	TCS Research	Hyderabad, India
	Sequential decision making, stochastic linear bandits, Thompson sampling, regret analysis under Dr. Sanjay P. Bhat	
	Project Associate	Jun 2019 – Jul 2019
	IIT Tirupati	Tirupati, India
	Research Intern	May 2018 – Jul 2018
	Lekha Wireless Solutions	Bengaluru, India
Teaching	Teaching Assistant	University of Illinois Urbana–Champaign
	ECE 434: Real-World Algorithms for IoT and Data Science <i>Office hours, grading, and student support</i>	Spring 2024, Spring 2025
Skills	Generative Modeling: Diffusion Models, NeRFs, Flow Matching, VAEs Frameworks: PyTorch, TensorFlow, HuggingFace Diffusers/Transformers Tooling: Slurm, Weights & Biases, Git, Streamlit, $\text{\LaTeX}$ Programming: Python, C/C++, MATLAB, SQL	
Selected Coursework	ECE 598: Deep Generative Models	
	ECE 598: Generative Modeling with Diffusion and Flow Matching Models	
	CS 598: Generative Models for Audio	
	ECE 549: Computer Vision	
	ECE 543: Statistical Learning Theory	
	ECE 586: MDPs and Reinforcement Learning	
	ECE 534: Random Processes	
	ECE 490: Introduction to Optimization	
Thesis	[1] M.S. Thesis	
	Chaitanya Amballa	
	<i>M.S. in Electrical and Computer Engineering, UIUC, 2025</i>	
	[2] B.Tech. Thesis	
	Chaitanya Amballa	
	<i>B.Tech. in Electrical Engineering, IIT Tirupati, 2019</i>	
Leadership	House Captain, Team “Sivaliks”	IIT Tirupati, 2016–2017
	Member, Technical Committee	IIT Tirupati, 2017–2019
	Electrical Engineering Department Sports Representative	IIT Tirupati, 2017–2018